

NITAP



NATIONAL INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PRESENTATION ON

“EMAIL / SMS SPAM CLASSIFIER”

GUIDED BY:

D. SAMEERA MAM

PRESENTING BY:

K. SOUJANYA (422163)

D. AMULYA (422136)

FARIZ SAMEER (422146)

CONTENTS

- Abstract
- Technologies
- Libraries
- Machine Learning
- Data-set
- Problem Definition
- Description of Dataset
- Methodolgy
- Algorithms
- Conclusion
- References

ABSTRACT

In today's globalized world, email / SMS is a primary source of communication. This communication can vary from personal, business, corporate to government. With the rapid increase in email usage, there has also been increase in the SPAM emails. SPAM emails, also known as junk email involves nearly identical messages sent to numerous recipients by email. Apart from being annoying, spam emails can also pose a security threat to computer system. It is estimated that spam cost businesses on the order of \$100 billion in 2007. In this project, we use text mining to perform automatic spam filtering to use emails effectively. We try to identify patterns using classification algorithms of Machine Learning to enable us classify the emails as HAM or SPAM

TECHNOLOGIES

Technologies used:

Machine Learning algorithms

Python

Python :- Python is an interpreted, object oriented , high level programming language with dynamic semantics.

Libraries:

Pandas

Numpy

Sklearn

NLTK

LIBRARIES

Pandas:- Pandas is one of the tools in Machine Learning which is used for data cleaning and analysis. It has features which are used for exploring, cleaning, transforming and visualizing from data

NumPy:- NumPy is a library for the Python programming language, adding support for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions to operate on these arrays. Moreover, NumPy forms the foundation of the Machine Learning stack.

NLTK:- NLTK is intended to support research and teaching in NLP or closely related areas, including empirical linguistics, cognitive science, artificial intelligence, information retrieval, and machine learning.

Matplotlib:- Matplotlib is a low-level library of Python which is used for data visualization. It is easy to use and emulates MATLAB like graphs and visualization. This library is built on the top of NumPy arrays and consist of several plots like line chart, bar chart, histogram, etc.

MACHINE LEARNING

- What is Machine Learning?
 - Machine Learning is a branch of artificial intelligence that develops algorithms by learning the hidden patterns of the datasets used it to make predictions on new similar type data, without being explicitly programmed for each task.
 - ML can find patterns and insights in large datasets that might be difficult for humans to discover.

DATA SET

- A machine learning dataset is a collection of data that is used to train the model. A dataset acts as an example to teach the machine learning algorithm how to make predictions. dataset as "a collection of data that is treated as a single unit by a computer". This means that a dataset contains a lot of separate pieces of data but can be used to train an algorithm with the goal of finding predictable patterns inside the whole dataset.

- How to train the data?

AI training data will vary depending on whether you're using supervised or unsupervised learning. Unsupervised learning uses unlabeled data. Models are tasked with finding patterns (or similarities and deviations) in the data to make inferences and reach conclusions.

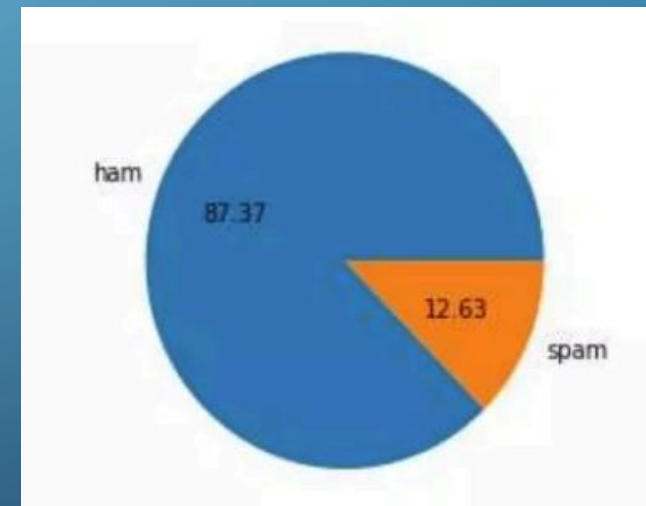
With supervised learning, on the other hand, humans must tag, label, or annotate the data to their criteria, in order to train the model to reach the desired conclusion (output). Labeled data is shown in the examples above, where the desired outputs are predetermined.

PROBLEM DEFINITION

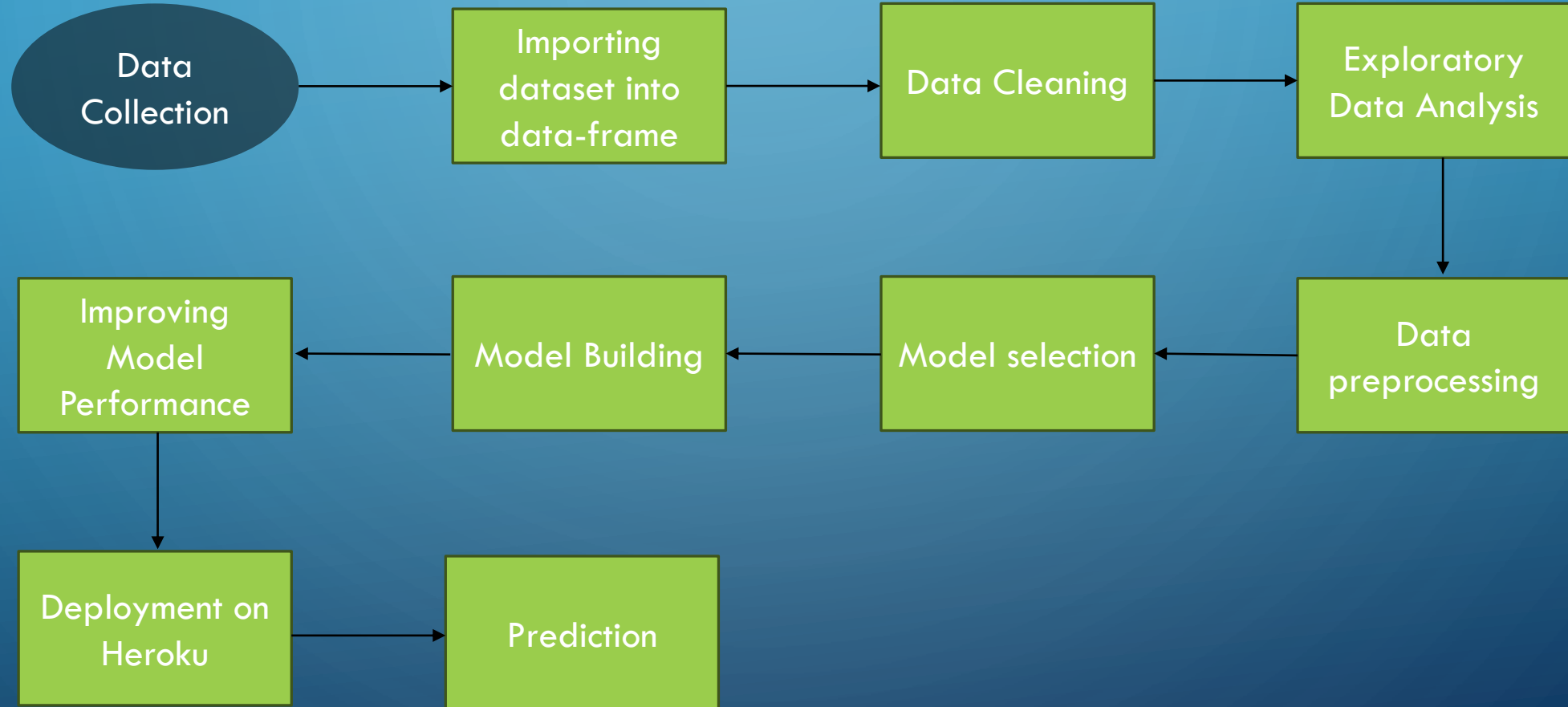
- Short Message (SMS) and email has grown into a multi-billion dollars commercial industry.
- SMS spam is still not as common as email spam.
- SMS Spam is showing growth, and in 2012 in parts of Asia up to 30% of text messages was spam.

DESCRIPTION OF DATASET

	target	text
2413	ham	Lol please do. Actually send a pic of yourself...
4234	spam	FREEMSG: Our records indicate you may be entit...
708	spam	To review and KEEP the fantastic Nokia N-Gage ...
3062	spam	Hi babe its Jordan, how r u? Im home from abro...
3881	ham	Short But Cute: \Be a good person



METHODOLOGY



ALGORITHMS

- Different algorithms used for email spam detection:-
 - 1. Deep learning
 - 2. Naive Bayes
 - 3. Support Vector Machines
 - 4. K-Nearest Neighbour
 - 5. Rough Sets
 - 6. Random Forests
 - 7. Multinomial naive

CLASSIFICATION OF ALGORITHMS (NAÏVE BAYES)

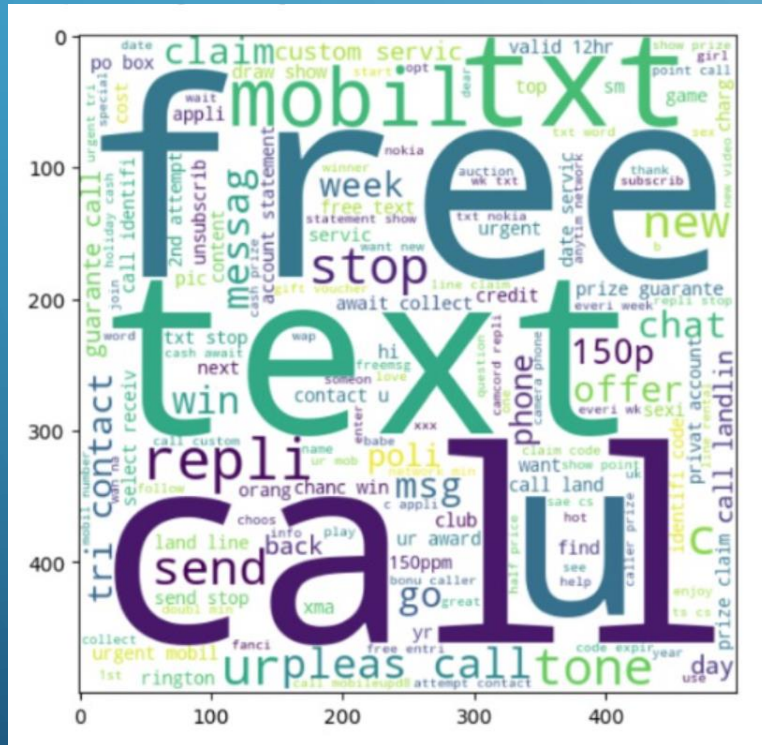
- NB algorithm is applied to the final extracted features. The accuracy and precision of this algorithm makes it a desirable classifier for spam detection problems. Applying naïve Bayes with multinomial event model to the dataset and using 10-fold cross validation results in Table.

```
[89]: performance_fd.merge(temp_fd,on='Algorithm')
```

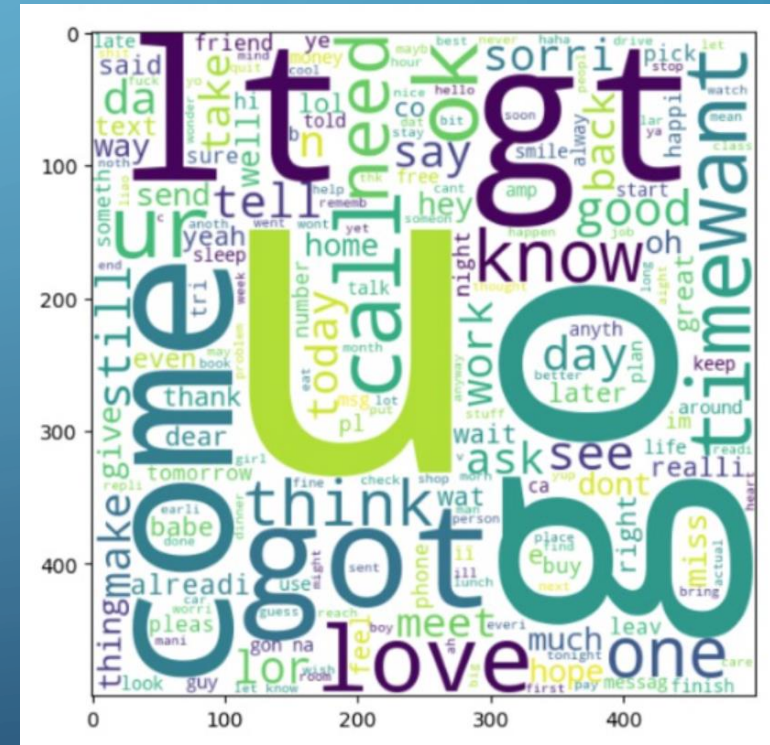
[89]:	Algorithm	Accuracy	Precision	Accuracy_max_ft_3000	Precision_max_ft_3000
0	KN	0.905222	1.000000	0.905222	1.000000
1	NB	0.970986	1.000000	0.970986	1.000000
2	RF	0.975822	0.982906	0.975822	0.982906
3	SVC	0.975822	0.974790	0.975822	0.974790
4	ETC	0.974855	0.974576	0.974855	0.974576
5	LR	0.958414	0.970297	0.958414	0.970297
6	AdaBoost	0.960348	0.929204	0.960348	0.929204
7	xgb	0.967118	0.926230	0.967118	0.926230
8	GBDT	0.946809	0.919192	0.946809	0.919192
9	BgC	0.958414	0.868217	0.958414	0.868217
10	DT	0.932302	0.833333	0.932302	0.833333

WORD CLOUD FOR SPAM / HAM WORDS

Using matplotlib:



Top spam words



Top ham words

WHICH EMAIL / SMS IS GENERALLY LONGER?

- Here, we have calculated average word count for Ham Emails and Spam Emails separately and then predicted which emails are generally longer.

- Average Word Count for Ham Emails: 4516.00 words

- Average Word Count for Spam Emails: 653.000 words

So, it can be concluded that, Ham emails are generally longer than spam emails.

	num_alphabets	num_words	num_sentences
count	4516.000000	4516.000000	4516.000000
mean	70.459256	17.123782	1.820195
std	56.358207	13.493970	1.383657

Ham average

	num_alphabets	num_words	num_sentences
count	653.000000	653.000000	653.000000
mean	137.891271	27.667688	2.970904
std	30.137753	7.008418	1.488425

Spam average

CONCLUSION

- Spam is a major problem in today's world. Spam messages are the most unwanted messages the end user clients receive in our daily lives. Spam emails are available nothing but an ad for any company, any kind of virus etc. It will be too much. It is easy for hackers to access our system using these spam emails.

REFERENCES

<https://www.geeksforgeeks.org/data-cleansing-introduction/amp/>

<https://www.geeksforgeeks.org/ml-handling-missing-values/amp/>

<https://www.geeksforgeeks.org/what-is-exploratory-data-analysis/>

<https://www.geeksforgeeks.org/handling-inconsistent-data/amp/>

<https://aiswaryasivakumar8.medium.com/creating-your-own-model-architecture-in-machine-learning-a-step-by-step-guide-522236c340d2>

- For dataset.
 - <https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset>

The image features a dark blue gradient background. In the corners, there are decorative white line art elements resembling circuit boards or neural networks, with lines and small circles connecting them. The text "THANK YOU" is centered in a large, bold, white sans-serif font.

THANK YOU